Geometry of asymmetric reciprocal CNV effects on height and BMI

Diagonal versus axis-aligned displacement, and what each implies for genome-wide burden

Summary

Reciprocal copy-number variants displace quantitative traits in a partly mirrored fashion: deletion and duplication effects are negatively correlated, with duplication effects the weaker of the two. A mirror model alone cannot account for the observed loci, and predicts the wrong direction for genome-wide duplication burden. Directional asymmetry is required. This report asks what geometry that asymmetry takes.

The previously published models displace every affected locus by the same vector, with the deletion and duplication components both non-zero, so each locus moves diagonally. We add an axis-aligned alternative in which each affected locus is displaced along one axis only, deletion-side with probability w and duplication-side with probability 1-w. The aggregate cloud still drifts diagonally, but no individual locus does. We also add a reparameterised model that estimates only the aggregate per-gene drift, dropping an event-frequency parameter that the data cannot identify.

Principal findings

- For height the decisive constraint turned out to be the published one. Restricting the shift to the lower-left quadrant forces the fitted angle onto its 85 degree boundary in every model. Releasing that bound moves the estimate to 119 degrees, past 90, where the deletion effect is attenuated toward the null while the duplication effect is driven past it. That is buffering in the sense of Sacks et al., and it is not representable at any parameter value inside the published window.
- The sparse model with buffering takes every likelihood-based criterion, with three parameters rather than the four the axis-aligned model needs: best conditional AIC, BIC and leave-one-locus-out, the widest posterior predictive margin of any asymmetry model, and a predicted duplication burden of -0.0744 cm per Mb against an observed -0.0784, closer than any other model.
- It does not win everything. The buffered model implies that duplication effect decreases with gene count, because a large duplication displacement combined with rising occupancy overwhelms the positive trend contributed by the mirror component. The observed trend is positive and nominally significant. On that one diagnostic the axis-aligned sparse model is the best fit and buffering falls outside the observed interval. No single height model is therefore consistent with every line of evidence, and the two should be reported together.
- All three lower-left-constrained models reproduce the duplication mean but break the deletion-duplication correlation, a mechanical consequence of a shared diagonal shift inflating the covariance. Buffering relieves that, because its two components have opposite sign and the induced covariance is negative rather than positive.
- For BMI the event frequency is not identified: the profile likelihood is flat and q runs to whatever upper bound it is given. Estimating only the identified aggregate drift uses two parameters instead of three and wins both conditional AIC and BIC, resolving a criterion conflict present in the earlier analysis.
- The two traits therefore differ in mechanism. Height requires stochastic, discrete occupancy: a mean-only model is worse by more than 31 AIC units. BMI does not; a deterministic drift proportional to gene count is sufficient and preferred.
- Every asymmetry model reverses the incorrect duplication-burden direction predicted by the mirror model alone. For height the axis-aligned sparse model predicts the strongest adverse duplication

Methods

Data

Single-locus analyses used 33 independent reciprocal height loci (21 mirror, 12 asymmetric) and 32 reciprocal BMI loci, taken from the exact Sacks et al. supplementary statistics. The overlapping 22q11.2 A-D interval was excluded. Each locus contributes a deletion and duplication beta, their standard errors, and an annotated gene count. Genome-wide burden coefficients are taken from Macé et al.; the overall burden direction is taken from Auwerx et al.

Mirror component

A CNV effect is modelled as the additive sum of latent per-gene effects. Kosmicki et al. informs the height sign prior (70 per cent of deletion or loss-of-function effects decrease height); Akbari et al. informs the BMI sign prior (posterior mean 78.6 per cent of predicted loss-of-function effects increase BMI). Contributor frequency, mean absolute per-gene effect, duplication-to-deletion magnitude ratio and residual dispersion were estimated from the Sacks CNVs themselves rather than imported from ascertainment-selected exome effects. All asymmetry models below are fitted conditional on this calibrated mirror component, so their information criteria are directly comparable.

Asymmetry geometries

Diagonal models displace an affected locus by b(cos theta, sin theta) with theta constrained to the interval 5 to 85 degrees, so both components are necessarily non-zero. Trait sign confines the displacement to the lower-left quadrant for height and the upper-right for BMI. Two occupancy structures are used: a continuous process in which many small effects accumulate with gene count, and a sparse process in which the presence of at least one rare gene produces a fixed displacement with probability 1-(1-q)^G.

The 5 to 85 degree window is itself a modelling choice, and it is the one that mattered. It forces both components of the shift to share a sign, so an affected locus must be displaced further from the null in both dosage states. Widening the window to 5 to 175 degrees keeps the duplication component pointing toward the asymmetric direction but lets the deletion component take either sign. Beyond 90 degrees the fitted displacement attenuates the deletion effect toward the null while carrying the duplication effect past it, which is buffering as defined in Sacks et al.: the locus is asymmetric because the duplication side carries the effect and the deletion side is buffered, not because both dosage states are damaging. The buffered models take the same three parameters as the models they generalise, and the published models are the restricted case.

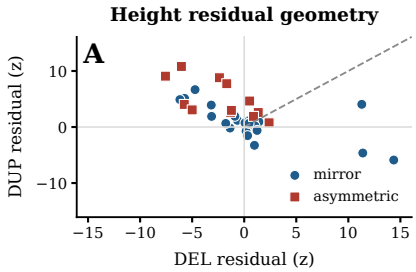
Axis-aligned models replace the shared vector with a mixture. An affected locus is displaced by (b_DEL, 0) with probability w, or by (0, b_DUP) with probability 1-w. Both occupancy structures are implemented, giving four parameters (q, w, b_DEL, b_DUP) against three for the diagonal models. The discriminating structure is second-moment: a shared diagonal shift adds a positive term to the deletion-duplication covariance, whereas axis-aligned events add none, because no single event moves both coordinates.

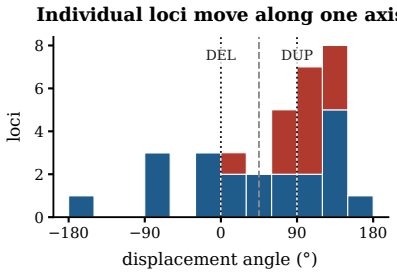
The aggregate model estimates only the quantity the data identify. Because the continuous likelihood is flat in q, with the product q*b nearly constant, this model drops q and estimates the per-gene drift directly as a deterministic displacement with no added dispersion, using two parameters. Note that this collapses the diagonal versus axis-aligned distinction, which survives only in the dispersion term.

Model comparison and checks

Models are compared by conditional AIC and BIC and by conditional leave-one-locus-out log predictive density, refitting the asymmetry parameters within each fold. Posterior predictive checks simulate 1200 replicate datasets per model and compare nine summary statistics with their observed values; a posterior predictive CDF below 0.025 or above 0.975 is treated as a failure. Each fitted gene-level model is propagated to expected deletion and duplication burden, calibrated to the Macé deletion coefficient by allowing an effective density of dosage-responsive genes per Mb, and the resulting duplication direction is compared with Auwerx.

Reproducibility



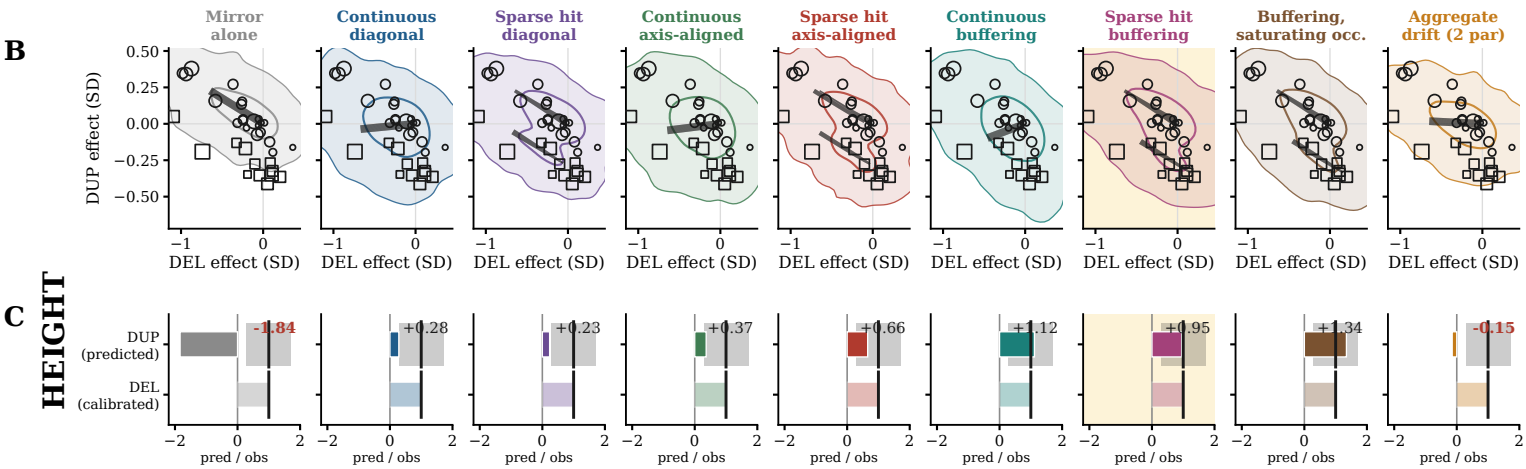


Published class vs displacement axis

	DUP-dom.	other
asymmetric	11	1
mirror	10	11

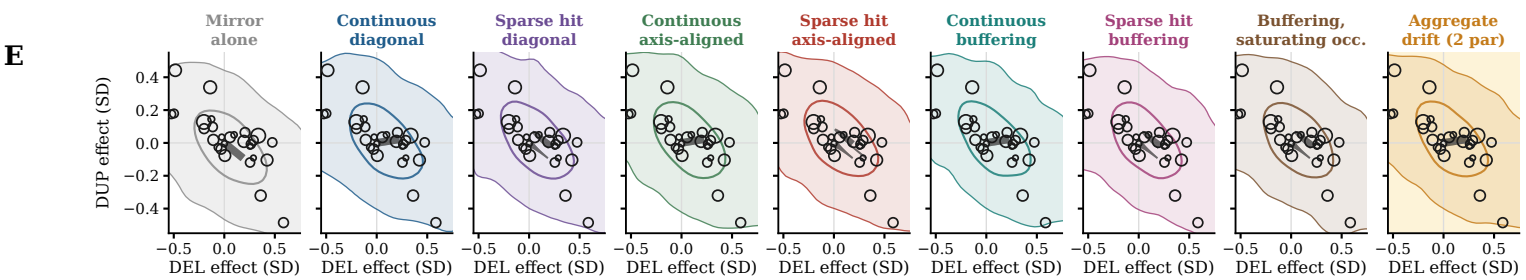
Fisher exact $P = 0.022$

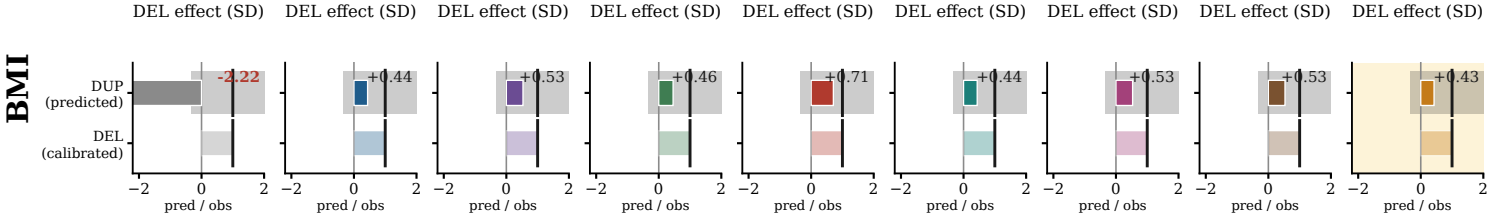
The published asymmetric class tracks the duplication axis; the three largest deletion-axis movers are labelled mirror.



D

	Mirror alone	Continuous diagonal	Sparse hit diagonal	Continuous axis-aligned	Sparse hit axis-aligned	Continuous buffering	Sparse hit buffering	Buffering, saturating occ.	Aggregate drift (2 par)
ΔAIC	77.2	7.7	5.3	8.7	4.6	2.6	**0.0**	1.8	35.9
ΔBIC	72.7	7.7	5.3	10.2	6.1	2.6	**0.0**	3.2	34.4
ΔLOO	38.8	2.7	1.9	2.2	1.5	0.8	**0.0**	0.5	21.7
PPC DUP mean	0.000	0.087	0.052	0.126	0.083	0.341	0.191	0.213	0.009
PPC DEL-DUP corr.	0.370	0.013	0.007	0.011	0.037	0.054	0.096	0.094	0.096





G

	Mirror alone	Continuous diagonal	Sparse hit diagonal	Continuous axis-aligned	Sparse hit axis-aligned	Continuous buffering	Sparse hit buffering	Buffering, saturating occ.	Aggregate drift (2 par)
ΔAIC	4.0	2.1	3.8	4.1	5.9	2.1	3.8	5.8	**0.0**
ΔBIC	1.1	3.5	5.3	7.0	8.8	3.5	5.3	8.7	**0.0**
ΔLOO	2.1	**0.0**	1.0	0.1	1.2	0.2	1.2	1.2	0.1
PPC DUP mean	0.846	0.333	0.402	0.341	0.368	0.356	0.452	0.408	0.349
PPC DEL-DUP corr.	0.152	0.102	0.115	0.099	0.113	0.108	0.107	0.113	0.107

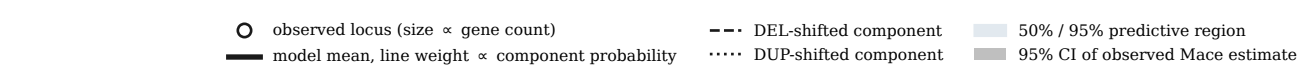


Figure legends

Main figure

Main figure. Geometry of asymmetric reciprocal CNV effects and its consequences for genome-wide burden.

- A. The motivating observation. Residuals of each height locus from the fitted mirror expectation, standardised by that locus's own standard error and signed so the asymmetric direction is positive. Individual loci concentrate near the deletion and duplication axes rather than on the 45 degree diagonal that the published models assume, and the published asymmetric class label tracks the duplication axis specifically.
- B, E. Reciprocal-effect plane for each model. Open symbols are the observed Sacks loci, sized by gene count; circles are mirror-class and squares asymmetric-class. Shading gives the 50 and 95 per cent predictive regions from 1200 simulated replicate datasets. Black lines are the model mean as a function of gene count, drawn as explicit mixture components with line weight proportional to component probability: one component for the mirror, continuous and aggregate models, two for the diagonal sparse model, and three for the axis-aligned sparse model, whose deletion-shifted and duplication-shifted branches are dashed and dotted.
- C, F. Burden propagation expressed as a multiple of the observed Macé estimate. The deletion row is the calibration anchor and is fixed at one by construction, so it is not a test; the duplication row is a free out-of-sample prediction. Grey band is the 95 per cent confidence interval of the observed duplication coefficient. Values left of zero have the wrong sign.
- D, G. Model comparison and posterior predictive checks, aligned under the model columns. Deltas are relative to the best model within each trait; posterior predictive values are CDFs, with failures below 0.025 or above 0.975 shaded red. The winning model is tinted, separately for each trait, because the winner differs between them.

- Note on B and E. Only the mirror and the sparse models keep the classic reciprocal mean trajectory, in which duplication effect moves opposite to deletion effect as gene count rises. The continuous-type models, including the aggregate drift preferred for BMI, add enough per-gene duplication drift to reverse that trend, so their mean curve tilts the same way on both axes. Their predicted deletion-duplication correlation nonetheless remains negative, because across loci that correlation is dominated by the residual covariance rather than by the gene-count trajectory. Figure 6 separates the two.

Observed burden coefficients are from Macé et al. 2017 Supplementary Table 1, which reports all four burden models the study tested. Standard errors are not published and have been reconstructed from the reported coefficient and P value assuming a two-sided Wald test; the intervals shown should be read with that in mind.

Table 1

Model comparison, posterior predictive behaviour and burden prediction, by trait and model. Fitted parameters, added parameter count, deltas against the best model within each trait, posterior predictive CDFs for the duplication mean and the deletion-duplication correlation, the effective dosage-gene density each model requires to match the observed deletion burden, and its resulting out-of-sample duplication prediction. The implied affected fraction applies to the sparse models only and is the mean over loci of 1-(1-q)^G.

Supporting figures

- Figure 1. Published reciprocal-CNV effects for both traits with standard errors, the fitted mirror model, and the direction of the residual asymmetry it leaves unexplained.
- Figure 2. Per-locus displacement geometry and the distribution of displacement angles, with the caveat that mirror-model residuals are structurally anti-correlated.
- Figure 3. Model comparison across all six models, three criteria and both traits.
- Figure 4. Simulated replicate datasets against observed height data for four representative models, with posterior predictive pass and fail marked.

Table 1. Model comparison, posterior predictive behaviour and burden prediction

All fits conditional on the calibrated mirror component. Burden columns are calibrated on the observed deletion coefficient, so the duplication prediction is out of sample.

Trait	Model	k	q	w_{DEL}	$shift_{DEL}$	$shift_{DUP}$	ΔAIC	ΔBIC	ΔLOO	PPC DUP mean	PPC corr.	genes per Mb	pred. DUP burden	obs. DUP burden	pred/ obs	implied affected
Height	Mirror alone	0	0.0000		0.0000	0.0000	77.23	72.74	38.75	0.000	0.370	4.61	+0.1440	-0.0784	-1.84	
Height	Continuous diagonal	3	0.0817		0.0049	0.0556	7.71	7.71	2.73	0.087	0.013	4.45	-0.0223	-0.0784	0.28	
Height	Sparse hit diagonal	3	0.0195		0.0246	0.2816	5.26	5.26	1.85	0.052	0.007	4.45	-0.0182	-0.0784	0.23	0.364
Height	Continuous axis-aligned	4	0.0811	0.000	0.0000	0.0578	8.65	10.15	2.20	0.126	0.011	4.61	-0.0286	-0.0784	0.37	
Height	Sparse hit axis-aligned	4	0.0682	0.486	0.0705	0.2884	4.62	6.11	1.52	0.083	0.037	4.14	-0.0514	-0.0784	0.66	0.727
Height	Continuous buffering	3	0.0764		-0.0420	0.0734	2.57	2.57	0.76	0.341	0.054	6.47	-0.0875	-0.0784	1.12	
Height	Sparse hit buffering	3	0.0191		-0.1954	0.3483	0.00	0.00	0.00	0.191	0.096	6.32	-0.0744	-0.0784	0.95	
Height	Buffering, saturating occ.	4	0.0270		-0.1916	0.3456	1.75	3.25	0.54	0.213	0.094	6.54	-0.1054	-0.0784	1.34	
Height	Aggregate drift (2 par)	2			0.0000	0.0036	35.91	34.42	21.66	0.009	0.096	4.61	+0.0115	-0.0784	-0.15	
BMI	Mirror alone	0	0.0000		0.0000	0.0000	4.04	1.11	2.13	0.846	0.152	9.31	-0.0701	+0.0315	-2.22	
BMI	Continuous diagonal	3	0.8000		0.0009	0.0025	2.08	3.54	0.00	0.333	0.102	7.61	+0.0137	+0.0315	0.44	
BMI	Sparse hit diagonal	3	0.0426		0.0254	0.0741	3.81	5.28	1.04	0.402	0.115	7.62	+0.0168	+0.0315	0.53	0.589
BMI	Continuous axis-aligned	4	0.8000	0.218	0.0039	0.0032	4.09	7.03	0.12	0.341	0.099	7.69	+0.0145	+0.0315	0.46	
BMI	Sparse hit axis-aligned	4	0.0422	0.111	0.1335	0.0881	5.88	8.81	1.16	0.368	0.113	8.25	+0.0224	+0.0315	0.71	0.586
BMI	Continuous buffering	3	0.8000		0.0009	0.0025	2.08	3.54	0.22	0.356	0.108	7.61	+0.0137	+0.0315	0.44	
BMI	Sparse hit buffering	3	0.0426		0.0254	0.0741	3.81	5.28	1.15	0.452	0.107	7.62	+0.0168	+0.0315	0.53	
BMI	Buffering, saturating occ.	4	0.0426		0.0254	0.0741	5.81	8.74	1.15	0.408	0.113	7.62	+0.0168	+0.0315	0.53	
BMI	Aggregate drift (2 par)	2			0.0008	0.0020	0.00	0.00	0.12	0.349	0.107	7.50	+0.0134	+0.0315	0.43	

k, added parameters. q, per-gene event frequency (not identified for the continuous BMI models). w_{DEL}, fraction of events on the deletion axis. Shifts in trait SD. Deltas are relative to the best model within each trait. PPC values are posterior predictive CDFs; below 0.025 or above 0.975 is a failure. Burden in cm/Mb (height) and kg/m^2/Mb (BMI); observed values from Mace et al. 2017 Supplementary Table 1. Implied affected fraction applies to the sparse models only.

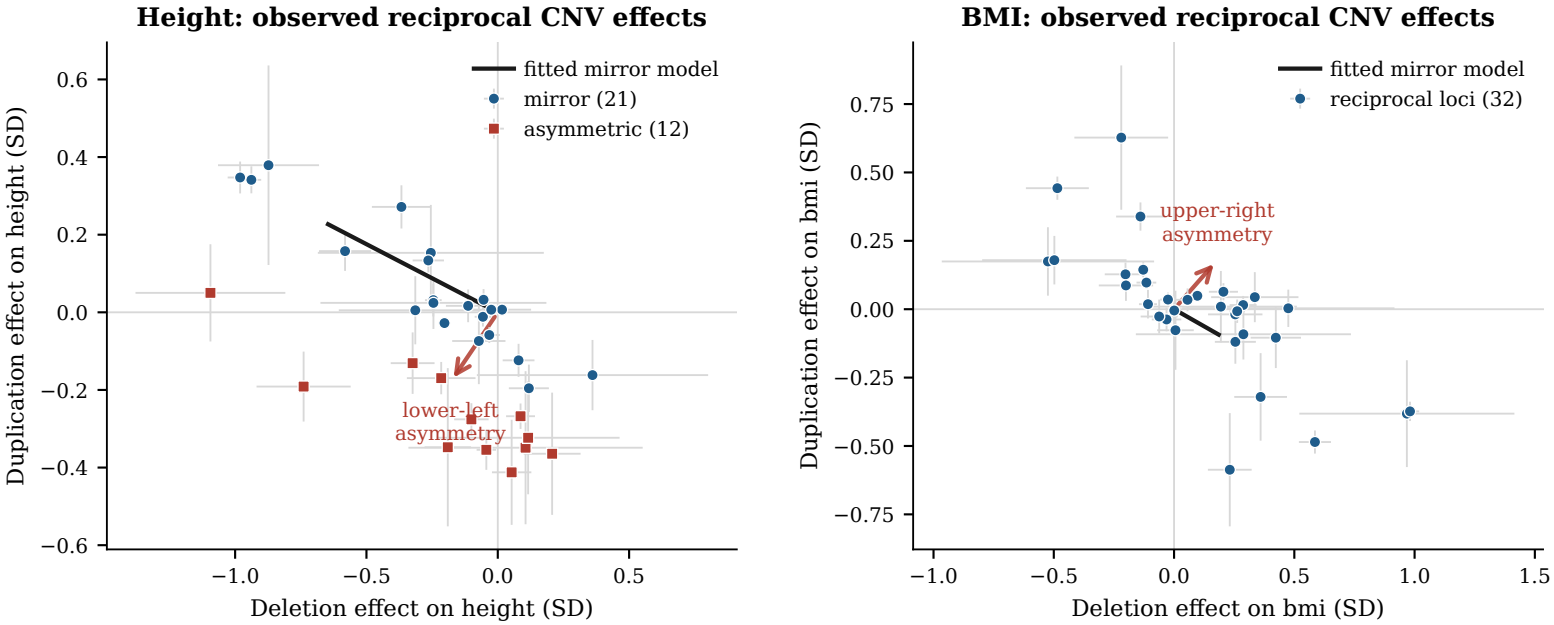


Figure 1. Published reciprocal-CNV effects. Each point is one locus, with standard errors. The fitted mirror model (black) captures the negative reciprocal trend in both traits. Height loci are labelled by their published dose-response class. The red arrow marks the direction

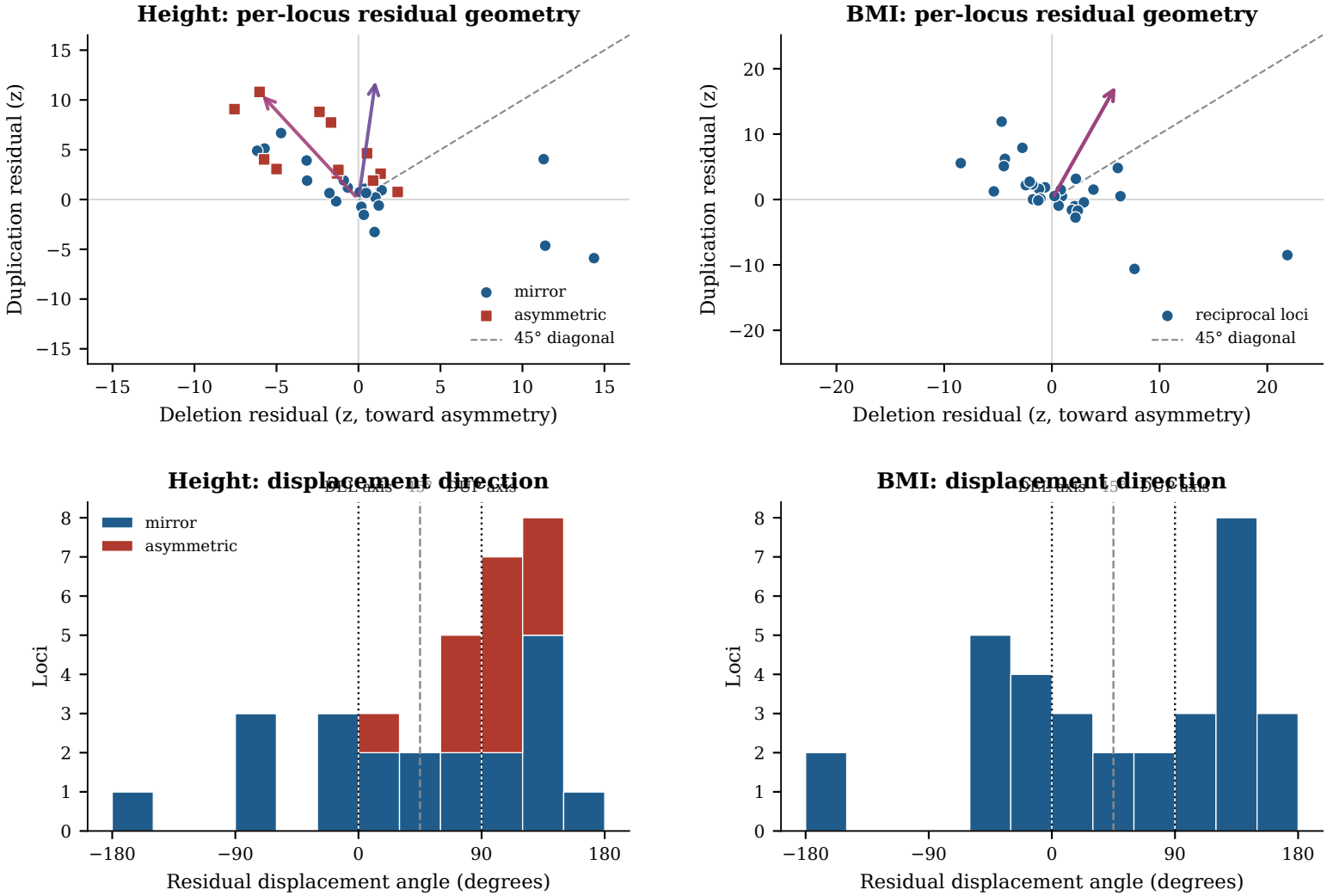


Figure 2. Per-locus displacement geometry. Top: residuals from the fitted mirror expectation, standardised by each locus's own standard error and sign-flipped so the asymmetric direction is positive on both axes, so a displacement drawn up and to the right here is lower-left

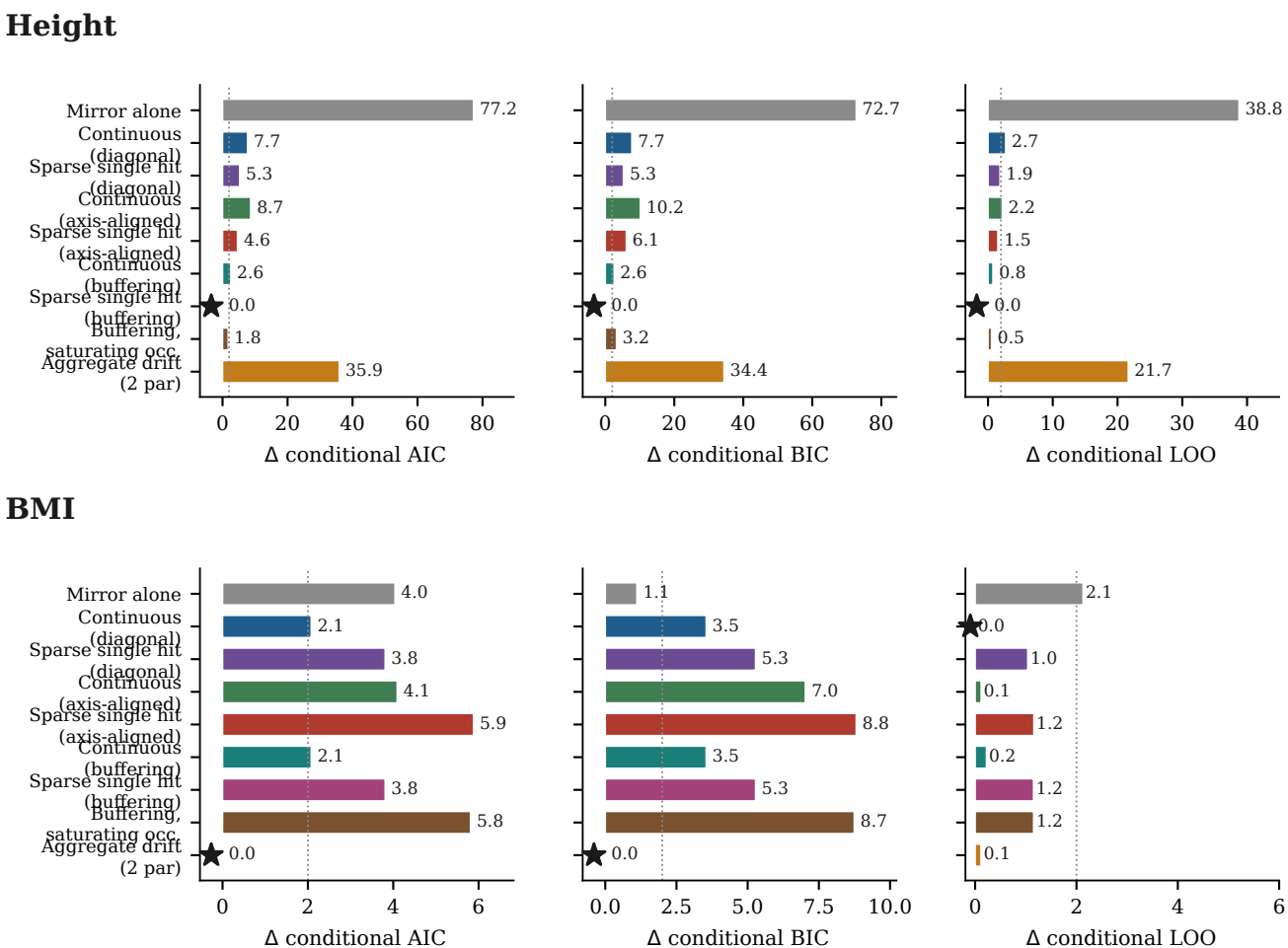


Figure 3. Model comparison, conditional on the calibrated mirror component. Bars show the penalty relative to the best model on each criterion; a star marks the winner and the dotted line marks the conventional two-unit threshold. Height prefers sparse, discrete

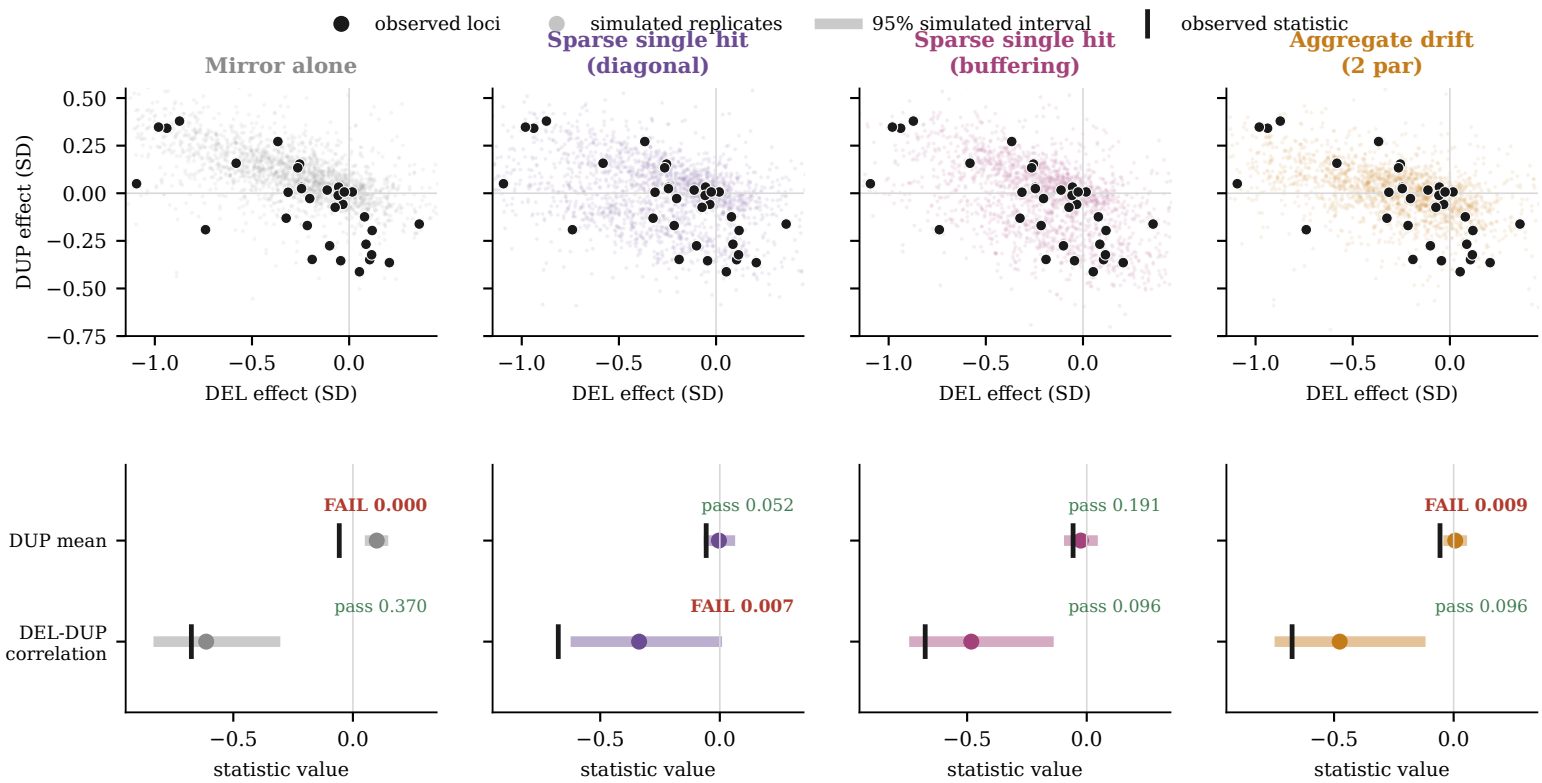


Figure 4. Simulations that do and do not reproduce the observed height data. Top: 60 simulated replicate datasets (pale) under each fitted model, with the observed loci overlaid (black). Bottom: two posterior predictive statistics, showing the 95 per cent simulated interval, the

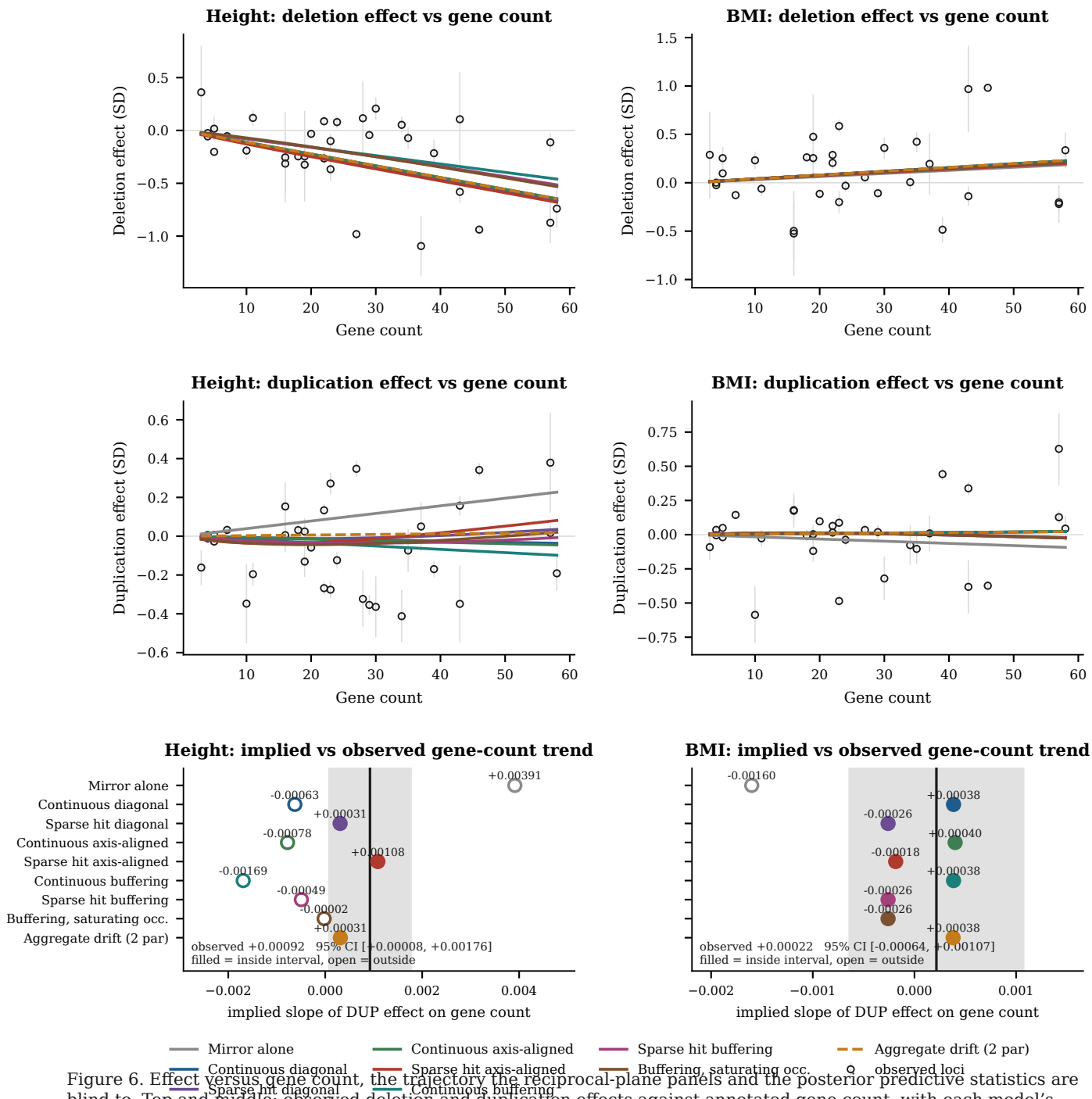


Figure 6. Effect versus gene count, the trajectory the reciprocal-plane panels and the posterior predictive statistics are blind to. Top and middle: observed deletion and duplication effects against annotated gene count, with each model's exact $E[\text{effect} \mid \text{gene count}]$ curve. These are evaluated rather than linearised, because the sparse models enter through

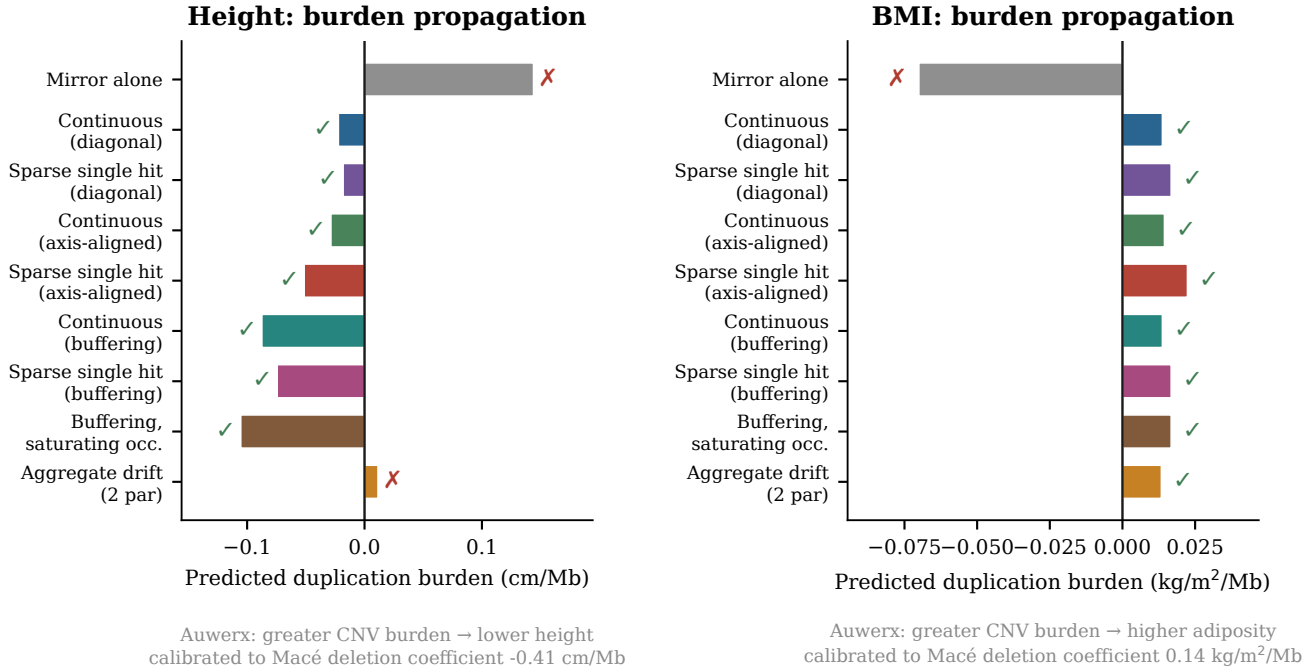


Figure 5. Propagation to genome-wide burden. Each fitted gene-level model is calibrated to the published Macé deletion coefficient by

Results

Height: single-locus fit

The mirror model alone is decisively rejected, trailing the best model by 77.2 AIC units. Among the models restricted to the lower-left quadrant, the axis-aligned sparse mixture is best on conditional AIC and leave-one-out, with a deletion-axis share of w = 0.486, close to an even split. But every one of those models sits on the 85 degree boundary of the permitted angular window, which is the signature of a constraint doing the work rather than the data.

Releasing the constraint resolves it. The sparse model with buffering settles at 119.3 degrees, with a deletion component of -0.1954 SD and a duplication component of +0.3483 SD. Applied at the median gene count this attenuates the expected deletion effect from about -0.28 to -0.08 SD while carrying the expected duplication effect from about +0.10 to -0.25 SD: the deletion side is buffered and the duplication side becomes the damaging one. It takes conditional AIC, BIC and leave-one-out simultaneously, ahead of the axis-aligned model by 4.62, 6.11 and 1.52 units, and ahead of the published diagonal sparse model by 5.26, 5.26 and 1.85. It does so with three parameters, one fewer than the axis-aligned model. The continuous buffered variant is second on every criterion, so the result is driven by the geometry rather than by the occupancy structure.

The aggregate model, which removes the dispersion contributed by stochastic occupancy, is worse by 35.9 AIC units and 21.7 units of predictive density, and its deletion component collapses to zero. Height asymmetry is therefore not a smooth drift; the discrete occupancy structure carries real information.

Height: posterior predictive behaviour

The mirror model reproduces the observed deletion-duplication correlation of -0.675 (CDF 0.370) but fails the duplication mean outright (CDF 0.000). Both published asymmetry models correct the mean and then fail the correlation (CDF 0.007 for the diagonal sparse model), which follows mechanically from a shared diagonal shift adding a positive term to the modelled covariance.

Buffering relieves this. Its two components carry opposite signs, so the covariance it induces is negative rather than positive, working with the observed reciprocal relationship instead of against it. The buffered sparse model reaches a duplication mean CDF of 0.191 and a correlation CDF of 0.096, the widest margin on both statistics of any asymmetry model. The axis-aligned model clears the same thresholds but only narrowly, at 0.083 and 0.037. Even so, the buffered simulated correlation of -0.483 still falls short of the observed value, so the correct statement is that buffering reduces the discrepancy that a same-signed shift creates rather than removing it.

BMI: identifiability and model choice

The continuous model is not identified for BMI. Along the profile, twice the log likelihood varies by only 0.34 units between q of 0.22 and q of 0.80, against a conventional threshold of 3.84, while the product of frequency and magnitude stays essentially constant. Raising the upper bound from 0.80 to 0.99 moves the estimate to the new bound and improves AIC by 0.03 units, leaving the model ranking unchanged. Only the aggregate drift is estimable.

Estimating that quantity directly, with two parameters rather than three, gives the best conditional AIC and the best conditional BIC, ahead of the continuous model by 2.08 and 3.54 units and ahead of the mirror model by 4.04 and 1.11. Its leave-one-out score is within 0.12 units of the best. The earlier conflict, in which AIC and leave-one-out favoured the continuous model while BIC favoured mirror alone, does not survive removing the unidentified parameter. The fitted per-gene drift is 0.0008 SD on the deletion axis and 0.0020 SD on the duplication axis, an upper-right direction consistent with the earlier estimate.

BMI posterior predictive checks discriminate nothing: every model, including mirror alone, lies inside the interval on all nine statistics. Nor does BMI provide an independent asymmetric class label, a size enrichment, or an occupancy result analogous to height. The diagonal and axis-aligned geometries differ by two AIC units but by only 0.12 units of predictive density, so the geometry of the BMI shift remains undetermined.

Results, continued

Genome-wide burden

Calibrated to the Macé deletion coefficient of -0.41 cm per deleted Mb, the mirror model predicts that duplication burden increases height by +0.1440 cm per Mb, which contradicts the adverse genome-wide direction reported by Auwerx. Every asymmetry model reverses the sign. The diagonal sparse model predicts -0.0182 and the axis-aligned sparse model -0.0514, roughly threefold larger. The aggregate model, already rejected on fit for height, also fails here, predicting +0.0115.

For BMI, calibrated to 0.14 kg/m2 per deleted Mb, the mirror model predicts a protective duplication burden of -0.0701. All asymmetry models move it to zero or positive; the preferred aggregate model predicts +0.0134, in the adverse direction reported by Auwerx. Burden therefore rejects the mirror model in both traits.

Because Macé et al. report a duplication-burden coefficient as well as a deletion one, the duplication comparison is quantitative rather than merely directional, and it does discriminate among the asymmetry models for height. Against an observed -0.0784 cm per Mb, the buffered sparse model predicts -0.0744, the axis-aligned model -0.0514 and the published diagonal sparse model -0.0182, the last being roughly fourfold too small. For BMI the observed duplication coefficient is not significant, so it remains a weak constraint there, and all asymmetry models are correctly signed and undershoot.

Independent evidence for discrete occupancy in height

Asymmetric-class height CNVs are longer than mirror-class CNVs (1.06 against 0.65 Mb, Welch p = 0.006) and contain more genes (31 against 22, p = 0.019), which is what a sparse occupancy process predicts and a within-class accumulation gradient does not. Consistent with that, the displacement score does not increase with size or gene count within either class (all p above 0.39).

Under the diagonal sparse model an event frequency of 1.95 per cent predicts an asymmetric prevalence of 36.40 per cent against an observed 36.36 per cent. Under the axis-aligned model the fitted frequency is higher and implies that 72.7 per cent of loci carry an event, which at first sight breaks this agreement. It does not: the axis model partitions that total into 35.3 per cent displaced along the deletion axis and 37.4 per cent along the duplication axis, each close to the observed prevalence. The published class label tracks the duplication axis, with 11 of 12 asymmetric-class loci displaced predominantly along it against 10 of 21 mirror-class loci (Fisher exact p = 0.022). The three largest deletion-axis movers are all labelled mirror. Under the axis model the prevalence match is evidence for one of two populations rather than for a single class.

Conclusions

- Mirror symmetry is a major component of both traits but cannot be the whole story. It is rejected by single-locus fit in height, by burden direction in both traits, and by the duplication mean in posterior predictive checking.
- Height asymmetry is sparse, discrete and stochastic. A deterministic drift is worse by more than 30 AIC units, so the occupancy structure carries real information.
- The lower-left constraint in the published analysis was the binding one. Every model restricted to that quadrant sits on the 85 degree boundary. Released, the estimate moves past 90 degrees to buffering: the deletion effect is attenuated toward the null while the duplication effect carries the trait. This is a correction to the published parameterisation, not merely an additional model.
- No single height model satisfies every line of evidence. Buffering wins conditional AIC, BIC and leave-one-out, both posterior predictive checks and the burden prediction; the axis-aligned mixture is the only model whose gene-count trajectory matches the observed trend. At 33 loci this should be reported as two competing descriptions rather than a settled winner, with the buffering result stated as the stronger of the two on likelihood grounds.
- Adding saturating occupancy, the natural reconciliation, does not resolve it. It very nearly closes the gene-count gap but loses on every information criterion, overshoots the observed duplication burden, and places its saturation point beyond the largest observed CNV. The tension between the